Hyperspectral Image Classification With Spatial–Spectral–Channel 3-D Attention and Channel Attention

Kun Sun, Fuxuan Dong, Wenxin Liu, Qing Wu, Xiaoming Sun, and Wei Wang

***Abstract*— For the advantages of focusing on important information and avoiding gradient vanishing, the attention mechanism (AM) and the densely connected network (DenseNet) are widely used in the hyperspectral image (HSI) classification. However, AMs applied to HSI classification (HSIC) do not consider different roles of each pixel, and the DenseNets ignore the different importance of different channel features, which affect HSIC performance. To solve the problems above, a novel HSIC method based on the spatial–spectral–channel 3-D AM (3DAM-SSC) and DenseNet improved with channel AM (CAM) is proposed, which is abbreviated as 3DAM-CAM DenseNet. First, the 3DAM-SSC is proposed and applied, the weight of each pixel point is determined by the pixel value itself and the correlations of different pixels in three dimensions. Second, the two-branch network structure with different scale convolutional kernels is used to fuse features. Third, a CAM is added after each DenseNet transition layer, and a total of three CAMs are used. Comparative experiments are conducted on hyperspectral datasets, which are Indian Pines (IP), KSC, and WHU-Hi-HongHu. The overall accuracy (OA) of 3DAM-CAM DenseNet in each dataset is 98.65%, 99.64%, and 98.73%, respectively, which are better than the compared methods.**

***Index Terms*— Channel attention mechanism (CAM), hyperspectral image classification (HSIC), spatial–spectral–channel 3-D attention mechanism (3DAM-SSC), two-branch network structure.**

I. Introduction

HIGH-RESOLUTION optical remote sensing images, characterized by submeter spatial resolution and rich spatial texture features, have been widely applied to remote sensing scene classification tasks, such as building classification [1], road traffic classification [2], and water body classification [3]. In order to solve the above problems, methods such as improving the model's semantic representation capability [4], [5] and cross-modal information integration [6] have been proposed. However, the limited spectral resolution inherent in high-resolution optical images poses challenges in distinguishing spectrally similar ground objects, such as various crop species. In contrast, hyperspectral images (HSIs), with high-spectral resolution and spatial–spectral coherence, provide discriminative feature information for precise ground object characterization, while clearly revealing spatial contextual relationships [7]. These advantages have enabled broad applications across diverse domains, including ecology [8], geology [9], urban land use [10], and agriculture [11]. HSI classification (HSIC), one of the fundamental tasks in HSI analysis, aims to categorize each pixel in an HSI into predefined land cover classes, including vegetation and water.

HSIC methods can be categorized into two main groups: machine learning methods and deep learning methods. Machine learning-based HSIC methods include the K-nearest neighbor (KNN) algorithm [12], random forest (RF) classifier [13], and the support vector machine (SVM) [14]. However, these approaches require manual feature engineering, demand high data volumes, and struggle with handling nonlinear relationships, and lack robustness in dealing with mixed noise scenarios involving Gaussian noise and sparse noise. Deep learning-based methods [15], [16], [17] enhance HSIC robustness by integrating denoising and classification into unified frameworks, leveraging joint optimization, hierarchical sparse modeling, and mixed-noise priors. Deep learning-based HSIC methods can be further categorized into unsupervised and supervised methods. Among the unsupervised methods, recent approaches aim at extracting informative representations from unlabeled data, which include clustering [18], unsupervised domain adaptation [19], and reinforcement-based learning [20] combined with deep learning.

Supervised HSIC methods can automatically learn high-level data features to effectively capture spectral, spatial, and joint spatial–spectral information in HSI. Chen et al. [21] were the first to apply deep learning to HSIC, using a stacked autoencoder (SAE) deep learning model to learn high-level features from HSI. Following this, researchers continuously improved the depth and complexity of deep learning-based HSIC networks to improve the performance of these methods. Hu et al. [22] introduced a weighted five-layer CNN, marking the first application of a multilayer CNN for spectral feature extraction (FE), with fully connected (FC) layers tasked with classifying extracted spectral features.

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Mou et al. [23] proposed a deep recurrent neural network that incorporates a multilayer 1D-CNN, effectively combining spectral correlation and band variability to capture deep features in the spectral dimension, thereby enhancing classification performance. To enhance the utilization of spatial information, Makantasis et al. [24] devised an HSIC method based on 2D-CNN. This approach employs a multilayer perceptron (MLP) for the final classification after initially extracting spatial features from pixels using a 2D-CNN. Fang et al. [25] introduced a compressed multibias network based on 2D-CNN (SMBN), which generates and adaptively selects response maps for different ground objects. Lee and Kwon [26] proposed a 2D-CNN-based HSIC approach designed to improve classification accuracy by augmenting the depth of the network and the quantity of convolutional kernels per layer. In [27], a deep FE method was explored to capture HSI spectral–spatial features using 3D-CNN for simultaneous extraction of spatial and spectral information. This method uses 3D-CNN to extract deep spectral and spatial information from the original HSI, while combining L_2 egularization and dropout to prevent overfitting and improve classification accuracy. Similarly, Sellami et al. [28] proposed a semi-supervised 3D-CNN for HSIC, where deep spectral and spatial features are obtained from both labeled and unlabeled training samples using an encoder–decoder framework, thereby enhancing classification performance. Xu et al. [29] achieved pixel-level classification by minimizing a fully convolutional network and a semi-supervised loss function. Roy et al. [30] and Ari [31] both proposed hybrid CNN networks combining 3D-CNN and 2D-CNN for HSIC, where the 3D-CNN part effectively extracts deep spectral–spatial features, while the 2D-CNN part helps reduce computational complexity and improve classification performance. Collectively, these methods improve FE capabilities and classification by progressively increasing the model's dimension and depth. However, this increase in dimension and depth also introduces the challenge of vanishing the gradient.

To address the vanishing gradient problem, Glorot et al. [32] proposed a normalized initialization method that ensures effective gradient propagation throughout the network by adjusting the initial weight distribution. This method helps to maintain the effectiveness of gradient propagation through the network. Glorot et al. [33] utilized the activation sparsity of the ReLU activation function [34], ensuring that the gradient is 1 in positive intervals and 0 in negative intervals. This approach mitigates the vanishing gradient issue within the network. He et al. [35] proposed the residual network (ResNet). Based on this, Zhong et al. [36] proposed the SSRN for HSIC. The skip connection structure in ResNet accelerates convergence and mitigates the gradient vanishing problem in deep neural networks as the depth increases. The skip connection structure inherent in ResNet accelerates convergence and mitigates the vanishing gradient issue when increasing the depth of the network. Inspired by SSRN, Wang et al. [37] proposed a fast dense spectral–spatial convolutional network for HSIC. This approach uses a densely connected network (DenseNet), where feature maps from preceding layers are accessible to all following layers, thereby enhancing feature reuse efficiency, and alleviating the gradient vanishing problem. Given that DenseNet can achieve superior performance compared with ResNet while utilizing fewer parameters and computational resources, DenseNet-based HSIC methods have emerged as a significant area of research. Enhancing the FE capability of DenseNet, Li et al. [38] proposed a method called MFDN. This method uses 2-D and 3-D convolutional layers to extract spatial and spectral features, respectively, and processes these features through a densely connected network to obtain more refined outputs. Xie et al. [39] proposed the multiscale DenseNet (MS-DenseNet), which integrates multiscale 3-D patches with DenseNet to extract features at varying scales across multiple levels. In addition, Cui et al. [40] and Ma et al. [41] incorporated an attention mechanism (AM) with densely connected networks to enhance the networks' FE capabilities. Collectively, these methods improve classification accuracy by strengthening FE. However, the front and back layers of DenseNet simply stitch together the shallow and deep feature maps by concatenating operation, without considering the interdependence between different channels.

For the advantages of focusing on important information, the AMs have been broadly adopted in HSIC presently. Sui et al. [42] effectively utilized the unique contributions of each band in HSIC by defining the weights of images across different bands. Zhao et al. [43] proposed a lightweight AM that generates weight vectors by compressing, capturing, and aggregating information from each spectral band to assign weights to images within each band, called a plug-and-play compact band weighting mechanism. Yu et al. [44] proposed a dense CNN incorporating a feedback AM, which captures the correlation between bands through a feedback AM, adjusts the weights of each band, and constrains the impact of different bands on classification performance. This approach constrains the impact of different bands on classification performance by capturing the correlations between bands and adjusting the weights assigned to each band. The above three methods are 1-D AMs that consider weights only from the spatial dimension and do not consider the influence of spectral dimension features on classification. Woo et al. [45] proposed CBAM, and Zhu et al. [46] proposed RSSAN, an HSIC network, in conjunction with CBAM. This AM sequentially computes attention maps across two separate dimensions, spatial and channel, and applies these maps to the input feature maps for adaptive feature adjustment. Guo et al. [47] proposed the dual-view spatial and spectral feature fusion (DSGSF) network, which emphasizes both spatial and spectral features. This network leverages spatial and channel AMs to enhance the representation of global spatial features, while employing a spectral AM to improve the spectral features across different views. Li et al. [48] introducted an HSIC method called DBDA, which employs a dual-branch dual-AM. This AM is characterized by a dual-branch framework consisting of a spectral branch and a spatial branch, in which channel attention and spatial AMs are employed to weight the spatial and spectral information, respectively. Roy et al. [49] introduced a method

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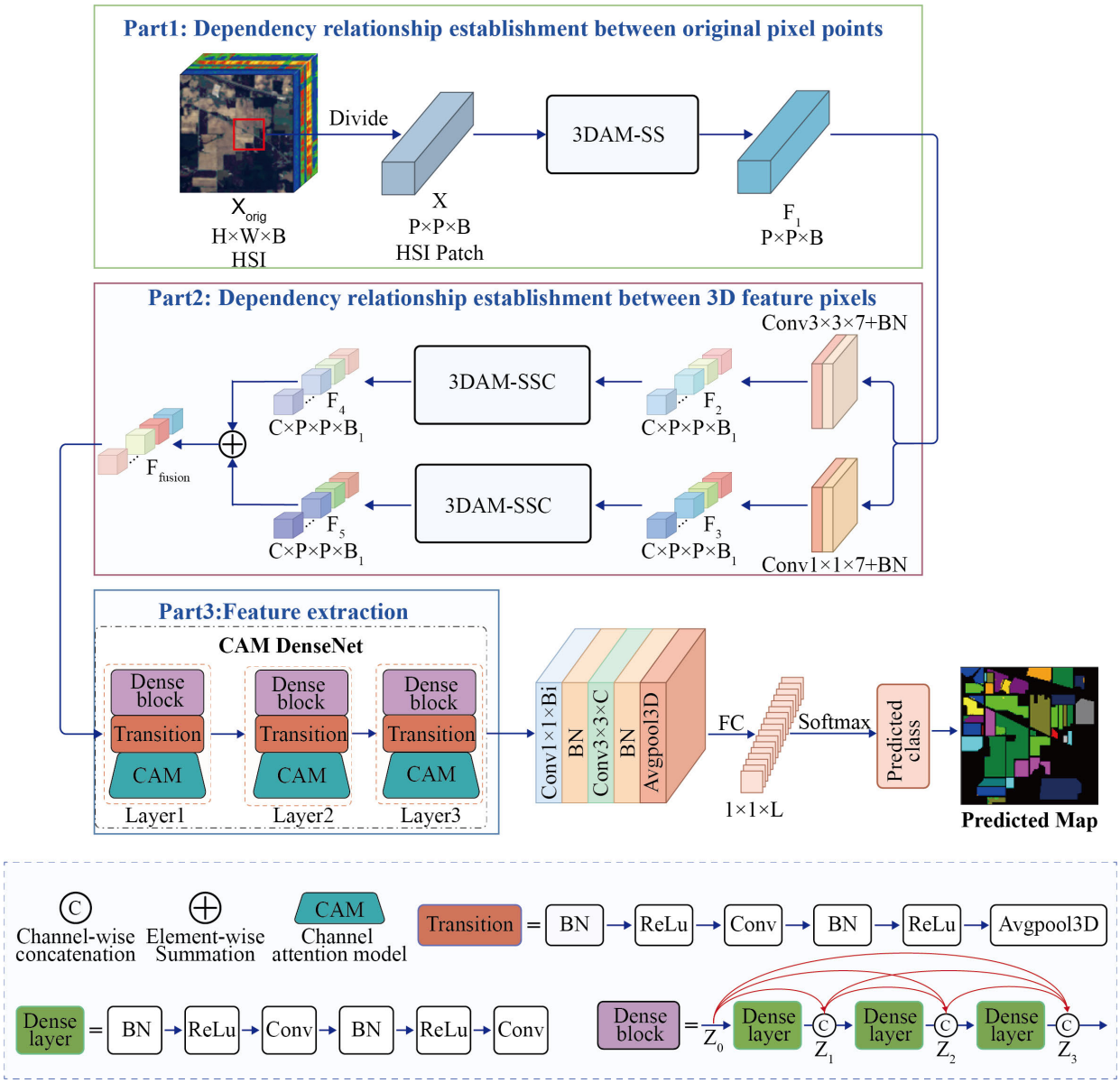


Fig. 1. Overall framework of 3DAM-CAM DenseNet. First, HSI is divided into 3-D patches X and processed through 3DAM-SS to establish the correlation between spectral and spatial pixel information. Second, convolutional kernels of different sizes are combined with 3DAM-SSC to capture correlations among different spectral–spatial–channel features. Subsequently, CAM DenseNet is used for deep FE.

incorporating the adaptive AM A^2S^2K, which utilizes a parallel framework to handle spectral and spatial features. This approach includes three key processes: kernel division, fusion, and selection, aimed at extracting spectral–spatial feature maps while simultaneously modifying the network's RF. Wan et al. [50] also enhanced the focus on spectral–spatial features through an adaptive AM. In addition, Li et al. [51] introduced the AWA-SVSS AM, which incorporates adaptive weight addition, as well as the ED-FVSS AM that utilizes feature vector self-similarity. AWA-SVSS generates two different weight maps which are linearly fused to calibrate the spectral–spatial patch via Euclidean distance and cosine similarity metrics. ED-FVSS generates spectral–spatial feature attention maps using Euclidean distance to calibrate and enhance the representation of spectral–spatial features. The aforementioned mechanisms consist of 2-D attention frameworks that utilize both spatial and spectral characteristics. To take into account the features across various dimensions more thoroughly, a spatial–spectral–channel 3-D AM has additionally been introduced, building upon the previously mentioned spectral–spatial attention module. Qiao et al. [52] introduced a four-branch parallel structure for a 3-D AM that fuses the weighted features from various branches to construct features with distinct dimensional dependencies. Liu et al. [53] introduced a 3-D AM known as shuffling and frequency AM (SFAM), characterized by a multibranch structure founded on the principle of "deconstruction–reconstruction." The above two AMs are 3-D, focusing, respectively, on the spatial, spectral, and channel dimensions to reweight features within each dimension, thereby improving HSI classification performance. However, these mechanisms compute attention weights at the HSI block level, assigning equal importance to all pixels within

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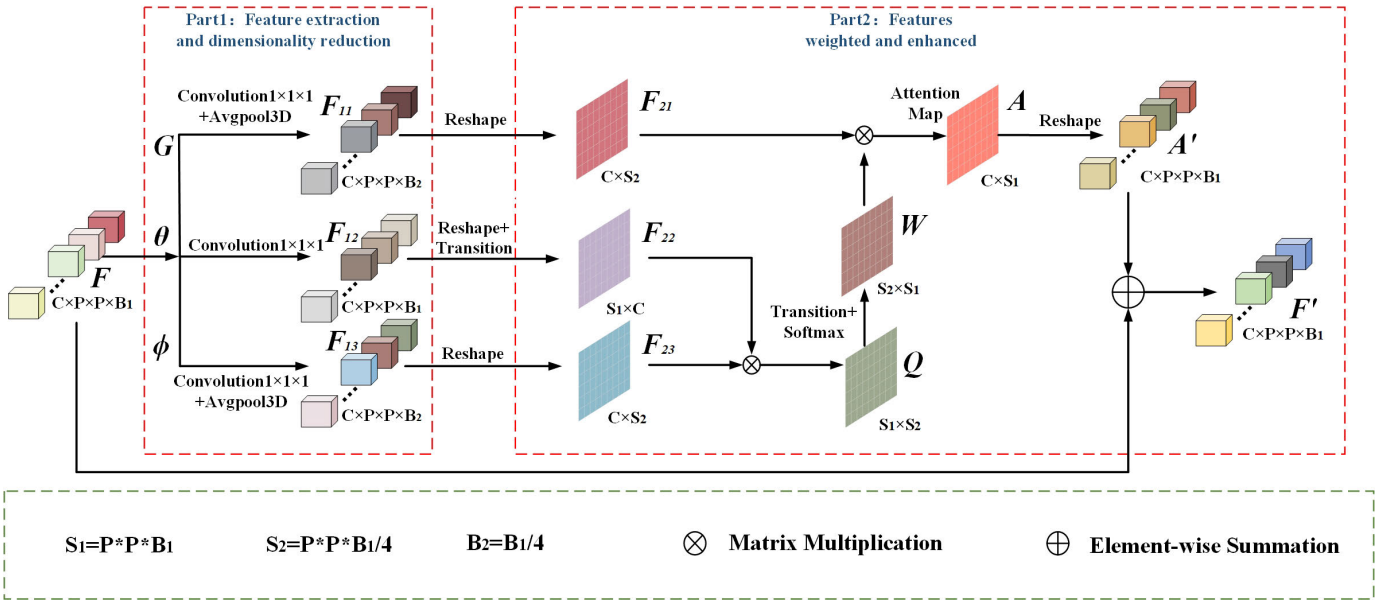


Fig. 2. Framework of 3DAM-SSC.

a block. This uniform weighting fails to capture the varying importance of individual pixels within the same image block.

To solve the two previously mentioned problems, drawing inspiration from nonlocal neural networks [54] and squeeze-and-excitation networks [55], we propose a pixel-level spatial–spectral–channel 3-D AM and an improved dense connectivity network capable of establishing channel feature dependencies, and on the basis of which, we propose a 3DAM-CAM DenseNet method for HSIC. The method combines spatial–spectral 3DAM-SS, spatial–spectral–channel 3DAM-SSC and the improved dense network CAM DenseNet. First, 3DAM-CAM DenseNet integrates image segmentation with 3DAM-SS to establish the dependency relationships among various pixels within the original pixel patch. Second, 3DAM-SSC is employed to construct features that capture spatial–spectral–channel pixel-level dependencies. A two-branch network structure is developed, utilizing different convolutional kernel sizes and feature fusion. Finally, a channel AM is implemented to enhance the densely connected network, thereby establishing dependencies between different channel features. The main contributions of this article are summarized as follows.

1) The 3DAM-SSC is proposed, which could establish the dependency relationships of different pixel points within an image block through establishing the correlation between the feature pixel information in three dimensions and the pixel values of different pixel points.
2) The CAM DenseNet is proposed, which could enhance the cross-channel correlation between features by adaptively recalibrating the channel features in DenseNet.
3) Establish the double-branch structure with convolutional kernels of different sizes for feature fusion, which could make the features more diverse. The 3DAM-CAM DenseNet is proposed by combining 3DAM-SSC, CAM DenseNet, and double-branch structure, which has superior performance on HSIC.

The rest of this article is structured as follows. Section II details the structure and principles of the 3DAM-CAM DenseNet network. Section III contains the experimental part and Section IV concludes this article.

II. Methodology

The 3DAM-CAM DenseNet method introduced in this article consists of three main parts, as illustrated in Fig. 1. In the first part, 3DAM-SS is proposed and utilized to process the HSI patch X to construct the dependencies between the original pixel points of the HSI patch X. In the second part, 3DAM-SSC is proposed and utilized to construct the dependencies between feature pixels in the three dimensions of spatial–spectral–channel, which is combined with a two-branch FE fusion structure based on different convolutional kernels to enhance the feature representation capabilities. In the third part, CAM is utilized to improve the original DenseNet by adding CAM after each transition layer of DenseNet to dynamically adjust the weights of different channel features and highlight the useful channel information, thereby improving the classification performance.

The original HSI dataset is denoted by $X_{\text{orig}} \in R^{H\times W\times B}$, H and W denote the height and width of the spatial dimension, respectively, and B denotes the number of spectral bands. Assume that the dataset X_{orig} contains N labeled pixels $x_{i,j} \in X_{\text{orig}}$, where $i = 1, \ldots, W$ and $j = 1, \ldots, H$, both have corresponding sets of one-hot label vectors $Y = [y_1, y_2, \ldots, y_N] \in \mathbb{R}^{1\times 1\times C}$, where C is the number of categories. To fully utilize the spectral and spatial information of the HSI, a new set of 3-D patch cubes $X = [x_1, x_2, \ldots, x_N] \in \mathbb{R}^{P\times P\times B}$ is formed by adjacent cubes centered on $x_{i,j}$, where P is the width of the patch cubes. In this article, each patch cube in X is used as an input to the model to classify its central pixel y_i.

A. 3DAM-SSC

To reflect the different importance of various pixel points within an image patch and to establish the interconnected

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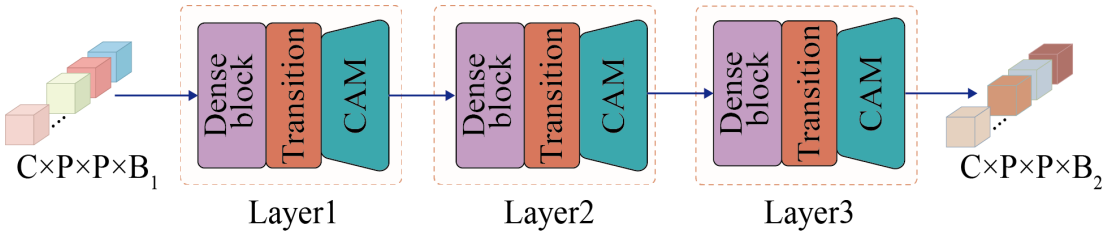


Fig. 3. Framework of CAM-DenseNet.

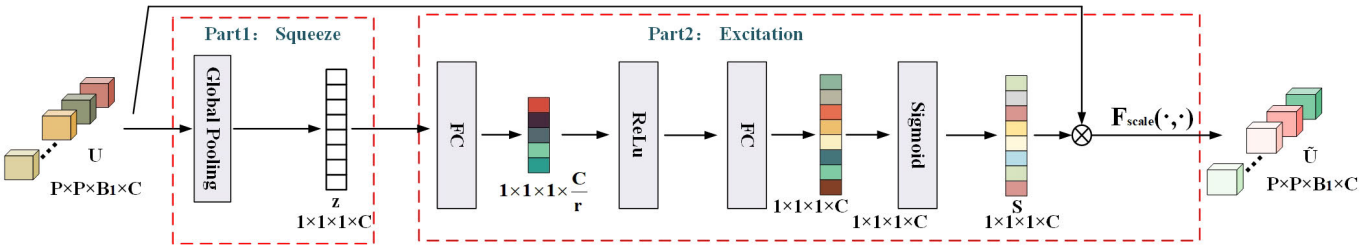


Fig. 4. Framework of CAM.

dependencies between pixels in the three dimensions of spatial–spectral–channel, this article proposes 3DAM-SSC by fusing the correlations between pixel information in these three dimensions and the pixel values of different pixel points, as shown in Fig. 2. 3DAM-SSC consists of the following two parts: FE and feature weighting.

1) Feature Extraction: This part uses $1 \times 1 \times 1$ convolution and 3-D average pooling to extract and downscale features F under different branches, $F \in \mathbb{R}^{C\times P\times P\times B_1}$, where C, P, and B_i are the number of channels, width, and spectral dimension, respectively. In Fig. 2, G, θ, and ϕ are the three branches used to capture the pixel correlation in different dimensions, and the feature F is processed by these branches to obtain the features F_{11}, F_{12}, and F_{13}, respectively, which can be, respectively, formulated as follows:

$$F_{11} = \text{Avgpool3D}\left(\sigma\left(W_G^l F + b_G^l\right)\right) \tag{1}$$

$$F_{12} = \sigma\left(W_\theta^l F + b_\theta^l\right) \tag{2}$$

$$F_{13} = \text{Avgpool3D}\left(\sigma\left(W_\phi^l F + b_\phi^l\right)\right) \tag{3}$$

where $*$ represents the convolutional operation, W_G^l, W_θ^l, and W_ϕ^l denote the parameters of the convolutional kernel for the three branches in the lth layer, b_G^l, b_θ^l, and b_ϕ^l are the biases of the three branches, and σ denotes the sigmoid activation function.

2) Feature Weighting and Feature Enhancement: This section employs the reshape operation, transition operation, matrix multiplication operation, and softmax operation to construct the normalized correlation matrix W for spatial–spectral–channel 3-D pixels. The attention map A is generated by multiplying the matrix W with F_{21}. In order to improve the original features, attention map A is integrated with the initial feature map using a residual framework. The process for computing the normalized correlation matrix W for spatial–spectral–channel 3-D pixels is conducted as follows:

$$W = \text{Softmax}\left((F_{22} \odot F_{23})^{\mathrm{T}}\right) \tag{4}$$

where $\odot$ is the vector dot product operation and T is the matrix transpose operation. The attention map A can be denoted by the following formula:

$$A = F_{21} \otimes W \tag{5}$$

where $\otimes$ is the matrix multiplication operation. With the above two steps, the dependencies between the pixels in the three dimensions can be established.

To enhance the pixel information at each pixel location, the feature attention map A is changed into an attention feature A' with the same shape as the input feature F after reshape operation, and then, A' is fused with the feature F at the pixel level to enhance the original input feature. The computation of the fusion of the attention map with the original feature map can be represented as follows:

$$F' = \text{Reshape}(A) \oplus F \tag{6}$$

where $\oplus$ is the elemental sum operation, $F' \in \mathbf{R}^{C\times P\times P\times B_1}$.

In order to construct the dependency between the original pixel points of HSI patch X, 3DAM-SS is a special application of 3DAM-SSC in focusing only on the spatial and spectral information, in Fig. 2, the value of the number of channels C is 1. This attentional mechanism weights different pixel points of the original data without channel information.

B. CAM DesneNet

To build the dependency relationships among various channel features, this article proposes CAM DenseNet, which means that a channel attention module [55] is added after each Transition layer of DenseNet, as shown in Fig. 3.

CAM DenseNet allows the output features of each layer to adjust the weights of different channel features by establishing a dependency between them and the channel features of the previous layer before the next propagation.

The CAM structure is illustrated in Fig. 4, and this module dynamically modifies the channel feature weights through two phases: squeeze and excitation.

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1) Squeeze: The squeeze operation generates a channel descriptor by aggregating feature mappings over the full channel dimension ($H \times W \times B_1$), using global average pooling generation, which is computed as follows:

$$z_c = F_{\text{sq}}(U_c) = \frac{1}{S \times S \times B_i} \sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{b=1}^{B} U_c(i, j, b) \tag{7}$$

where U_c is the cth channel feature, z_c is the channel descriptor mapped by the cth channel feature, and (i, j, b) is the position index of the channel feature pixel point.

2) Excitation: Excitation uses two nonlinear FC layers and the ReLU activation function to reduce the complexity of the model and enhance the generalization ability; the sigmoid function is used to calculate the weights of the features on different channels, denoted as follows:

$$\begin{aligned} s &= F_{\text{ex}}(z, W) = \sigma(g(z, W)) \\ &= \sigma(W_2 \delta(W_1 z)) \sum_{j=1}^{W} \sum_{b=1}^{B} U_c(i, j, b) \end{aligned} \tag{8}$$

where δ represents the ReLU [34]) activation function, $W_1 \in \mathbb{R}^{((C/m)\times C)}$ and $W_2 \in \mathbb{R}^{((C/m)\times C)}$.

The output feature $\tilde{U}_c$ is obtained by multiplying the input feature U with the weight s, which can be expressed as follows:

$$\tilde{U}_c = F_{\text{scale}}(U_c, s_c) = U_c \cdot s_c \tag{9}$$

where $\tilde{U} = [\tilde{u}_1, \tilde{u}_2, \ldots, \tilde{u}_c]$

III. Experimental Results

To assess the efficacy of the 3DAM-CAM DenseNet approach, this article performs experimental validation on three prominent HSI datasets: Indian Pines (IP), KSC, and WHU-Hi-HongHu (HH). This includes both comparative and ablation experiments. The experiments are conducted on a platform featuring an NVIDIA RTX 4090 24-GB GPU and a 16 vCPU AMD EPYC 9654 96-core processor, utilizing a Python 3.9 environment.

A. Introduction to the Dataset

The IP dataset was collected in June 1992 using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [56], covering a wavelength range from 400 to 2500 nm. This dataset comprises image spatial pixels measuring 145 × 145 and includes 200 bands that represent 16 land cover classes, such as maize, oats, wheat, and forests, with a total sample size of 10 249 individuals. In this article, the samples from the dataset are allocated across the training, validation, and testing sets in proportions of 10%, 10%, and 80%, respectively, with detailed counts provided in Table I.

KSC Dataset: This dataset was obtained in March 1996 using AVIRIS sensor imaging [56], with image space pixels of 512 × 614, 176 bands, containing 13 categories such as Oak, Scrub, and Wale, and the number of samples is 5211. This article designates the distribution of samples within the training, validation, and testing sets of this dataset as 10%, 10%, and 80%, respectively, with exact figures provided in Table II.

TABLE I

Number of Training, Validation, and Testing Samples in the IP Dataset, and the Colors of Different Sample Types

Class No.	color	Class Name	Train	Val	Test
1		Alfalfa	5	5	36
2		Corn-notill	143	143	1142
3		Corn-mintill	83	83	664
4		Corn	24	24	189
5		Grass-pasture	49	49	385
6		Grass-trees	73	73	584
7		Grass-pasture-mowed	3	3	22
8		Hay-windowed	48	48	382
9		Oats	2	2	16
10		Soybean-notill	98	98	778
11		Soybean-mintill	246	246	1963
12		Soybean-clean	60	60	473
13		Wheats	21	21	163
14		Woods	127	127	1011
15		Building-Grass-Trees-Drives	39	39	308
16		Stone-SteelTowers	10	10	73
		Total	1031	1031	8187

TABLE II

Number of Training, Validation, and Testing Samples in the KSC Dataset, and the Colors of Different Sample Types

Class No.	color	Class Name	Train	Val	Test
1		Scrub	77	77	607
2		Willow swamp	25	25	193
3		Cabbage palmhammock	26	26	204
4		Cabbage palm/oak hammock	26	26	200
5		Slash pine	17	17	127
6		Oak/broadleaf hammock	23	23	183
7		Hardwood swamp	11	11	83
8		Graminoid marsh	44	44	343
9		Spartina marsh	52	52	416
10		Cattail marsh	41	41	322
11		Salt marsh	42	42	375
12		Mud flats	51	51	401
13		Wate	93	93	741
		Total	522	522	4167

The WHU-Hi-HongHu (HH) dataset was collected from 16:23 to 17:37 on November 20, 2017 in Honghu, Hubei, China, covering a wavelength range of 400–1000 nm. This dataset includes 940 × 475 image spatial pixels, 270 bands, and 22 land cover categories such as Cotton, Rape, and Cabbage, with a total sample size of 386 693. In this article, the samples in the dataset are allocated to the training set, validation set, and testing set in proportions of 3%, 3%, and 94%, respectively. The detailed counts are shown in Table III.

B. Evaluation Indicators

This article evaluates the results by three metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient.

OA can represent the number of pixels correctly classified by the model as a percentage of the total pixels, and its

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TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES IN THE HH DATASET, AND THE COLORS OF DIFFERENT SAMPLE TYPES

Class No.	color	Class Name	Train	Val	Test
1		Red roof	421	421	13199
2		Road	105	105	3302
3		Bare soil	654	654	20513
4		Cotton	4898	4898	153489
5		Cotton firewood	186	186	5846
6		Rape	1336	1336	41885
7		Chinese cabbage	723	723	22657
8		Pakchoi	121	121	3812
9		Cabbage	324	324	10171
10		Tuber mustard	371	371	11652
11		Brassica parachinensis	330	330	10355
12		Brassica chinensis	268	268	8418
13		Small Brassica chinensis	675	675	21157
14		Lactuca sativa	220	220	6916
15		Celtuce	30	30	942
16		Film covered lettuce	217	217	6828
17		Romaine lettuce	90	90	2830
18		Carrot	96	96	3025
19		White radish	261	261	8190
20		Garlic sprout	104	104	3278
21		Broad bean	39	39	1250
22		Tree	121	121	3798
		Total	11590	11590	363513

calculation principle is as follows:

$$\mathrm{OA} = \frac{\mathrm{IM}}{\mathrm{IN}} \tag{10}$$

where IM denotes the correctly classified pixels and IN denotes the total number of pixels.

AA is calculated by summing the ratio between each correctly categorized pixel and the total number of that pixel, and then taking its mean value, and the following outlines the calculation principle:

$$\mathrm{AA} = \frac{\sum_{k=1}^{K} \frac{\mathrm{IM}_k}{\mathrm{IN}_k}}{K} \tag{11}$$

where IM_k represents the number of correctly classified kth feature labels, IN_k denotes the total number of kth feature labels, and K denotes the number of label categories.

Kappa is used to detect whether the classification result of the model is consistent with the correct result, and the calculation principle is as follows:

$$\mathrm{kappa} = \frac{\mathrm{IM} \times \mathrm{IN} - \sum_{k=1}^{K} \mathrm{IP}_k \times \mathrm{IN}_k}{\mathrm{IN} \times \mathrm{IN} - \sum_{k=1}^{K} \mathrm{IP}_k \times \mathrm{IN}_k} \tag{12}$$

where IP_k denotes the number of labels classified as the kth label by the model, and the characters IM, IN, and IN_k have the same meaning as above.

C. Experimental Parameter Settings

In this article, the adaptive nature of Adam's [58] adaptive learning rate optimizer is exploited to update the 3DAM-CAM DenseNet weight parameters for fast convergence of the parameters. The epochs are 200, the batch size is 32, the learning rate is set to 0.001, and the number of iterates is set to 5.

TABLE IV
OA (%) OF WEIGHT COEFFICIENTS UNDER DIFFERENT DATASETS

Weight Coefficient		OA (%)		
λ_1	λ_2	**IP**	**KSC**	**HH**
0.1	0.9	97.56	98.23	98.11
0.2	0.8	97.80	98.85	98.17
0.3	0.7	98.27	98.97	98.24
0.4	0.6	**98.48**	99.18	98.36
0.5	0.5	98.32	**99.39**	98.54
0.6	0.4	98.12	99.13	**98.73**
0.7	0.3	97.63	98.87	98.59
0.8	0.2	97.38	98.68	98.37
0.9	0.1	97.01	98.57	98.13

TABLE V
OA (%) OF 3DAM-CAM DENSENET USING DIFFERENT SPATIAL WINDOW SIZES

Patch Size	OA (%)		
	IP	KSC	HH
3×3	94.82	92.49	95.78
5×5	94.94	95.83	96.32
7×7	96.92	98.10	97.56
9×9	**98.65**	**99.64**	98.21
11×11	98.47	98.65	**98.73**
13×13	98.39	98.52	97.89
15×15	98.06	97.72	96.36

The second part of 3DAM-CAM DenseNet is a two-branch FE fusion structure with different convolutional kernels, and its fusion process can be summarized as follows:

$$F_{\mathrm{fusion}} = \lambda_1 \times F_4 + \lambda_2 \times F_5 \tag{13}$$

where the weights of the dual branches are λ_1, λ_2, $\lambda_1 + \lambda_2 = 1$.

To evaluate the import of varying weights in classification performance, this article conducts linear weighting experiments in two-branch feature fusion. The results of these experiments are presented in Table IV. Therefore, the two-branch feature fusion weights are $\lambda_1 = 0.4$ and $\lambda_2 = 0.6$ for the IP dataset, $\lambda_1 = 0.5$ and $\lambda_2 = 0.5$ for the KSC dataset, and $\lambda_1 = 0.6$ and $\lambda_2 = 0.4$ for the HH dataset.

To determine the appropriate number of convolutional kernels, we performed 3DAM-CAM DenseNet classification experiments utilizing various convolutional kernels in three datasets, with the results illustrated in Fig. 5. In this article, we have selected 24 as the number of convolutional kernels.

To set the patch size, this article uses the 3DAM-CAM DenseNet architecture to perform classification experiments using various patch sizes in three datasets, with the findings presented in Table V. According to Table V, the maximum OA value is achieved with a patch size of 9×9 for the IP and KSC datasets, while the highest OA value for the HH dataset occurs with a patch size of 11×11 . Consequently, in the following experiments of this article, the patch size will be configured to 9×9 for the IP and KSC datasets and 11×11 for the HH dataset.

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TABLE VI
OA, AA, AND KAPPA VALUES ON THE IP DATASET USING FIXED 10% OF TRAINING SAMPLES

Class	Methods									
#	ENL-FCN	A^{2}S^{2}K-ResNet	CVSSN	DSGSF	GMA-Net	LMSS-NAS	GSCViT	DSFNet	QuadNet	Proposed
1	**99.07±0.013**	95.81±0.031	85.95±0.017	95.37±0.070	**100.00±0.000**	28.07±0.000	97.26±0.023	79.23±0.173	86.92±0.149	96.48±0.025
2	96.17±0.020	98.73±0.009	97.68±0.043	97.58±0.010	76.45±0.000	84.82±0.091	91.96±0.091	97.42±0.005	98.11±0.011	**98.59±0.006**
3	97.58±0.011	98.49±0.006	**99.73±0.020**	94.24±0.060	63.73±0.000	98.43±0.0083	98.98±0.0083	98.48±0.009	99.03±0.006	98.93±0.004
4	96.94±0.002	98.42±0.011	95.36±0.042	93.17±0.080	**100.00±0.000**	78.11±0.017	98.36±0.012	99.48±0.004	96.63±0.017	97.75±0.017
5	99.06±0.006	99.20±0.006	97.22±0.160	95.77±0.05	79.53±0.000	93.03±0.000	**99.91±0.001**	95.70±0.045	96.92±0.028	99.24±0.017
6	98.31±0.002	99.77±0.006	**100.00±0.000**	99.17±0.011	98.60±0.000	86.88±0.037	**100.00±0.000**	94.03±0.074	97.96±0.020	99.66±0.037
7	**100.00±0.000**	96.15±0.003	92.47±0.016	91.03±0.070	**100.00±0.000**	28.57±0.000	95.06±0.0070	64.62±0.177	82.81±0.133	95.06±0.070
8	99.39±0.003	97.97±0.054	99.92±0.110	99.58±0.060	**100.00±0.000**	98.67±0.008	98.67±0.008	99.41±0.008	97.73±0.018	**99.91±0.001**
9	**100.00±0.000**	85.38±0.014	**100.00±0.000**	88.33±0.100	**100.00±0.000**	15.15±0.000	**100.00±0.000**	67.59±0.315	64.39±0.134	**100.00±0.000**
10	96.66±0.004	98.40±0.139	97.78±0.062	95.91±0.030	92.38±0.000	86.61±0.017	97.97±0.006	95.68±0.034	92.26±0.030	98.38±0.005
11	98.22±0.004	98.58±0.017	98.49±0.041	98.90±0.010	75.26±0.000	96.52±0.007	98.74±0.004	97.33±0.020	98.38±0.004	98.89±0.005
12	**97.76±0.016**	86.21±0.033	97.19±0.006	97.09±0.004	96.39±0.112	92.62±0.010	97.25±0.013	94.54±0.015	93.92±0.012	97.02±0.021
13	99.80±0.003	96.94±0.017	99.40±0.005	**100.00±0.000**	**100.00±0.000**	99.40±0.060	**100.00±0.000**	**100.00±0.000**	99.38±0.009	99.22±0.011
14	**99.77±0.005**	99.42±0.005	97.50±0.071	99.67±0.010	99.67±0.002	91.78±0.009	89.29±0.000	98.80±0.006	98.62±0.008	98.93±0.005
15	93.71±0.089	98.39±0.006	95.19±0.094	94.69±0.020	**100.00±0.000**	75.00±0.000	98.33±0.009	93.32±0.080	96.37±0.024	96.38±0.028
16	97.73±0.023	**99.54±0.006**	97.73±0.023	94.17±0.037	93.11±0.052	95.79±0.002	98.71±0.010	92.33±0.056	97.69±0.011	96.63±0.038
OA(%)	97.81±0.003	98.60±0.004	97.96±0.002	97.32±0.018	85.26±0.000	90.16±0.002	97.70±0.017	96.63±0.020	97.05±0.004	**98.65±0.002**
AA(%)	98.08±0.006	97.25±0.009	96.68±0.006	97.25±0.009	92.39±0.000	75.59±0.002	98.18±0.005	91.75±0.047	93.57±0.022	**98.19±0.006**
Kappa × 100	97.50±0.004	98.40±0.005	97.67±0.002	98.40±0.005	97.67±0.020	96.94±0.021	97.38±0.019	96.16±0.023	96.64±0.004	**98.46±0.003**

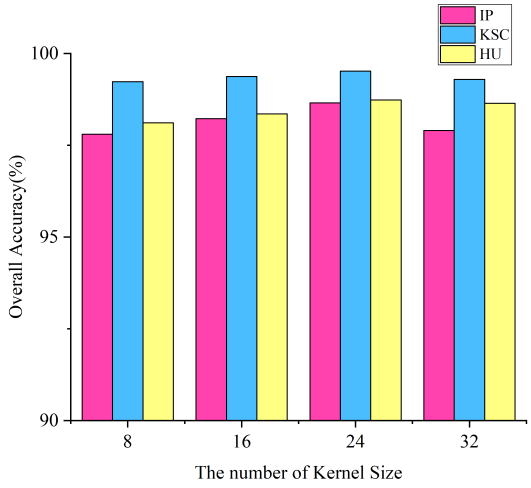


Fig. 5. OA of 3DAM-CAM DenseNet with different number of kernel size.

D. Comparison With the State of the Arts

In order to validate the effectiveness and classification performance of the proposed method, this section compares 3DAM CAM DenseNet with the state-of-the-art deep learning methods: ENL-FCN[59], A2S2K ResNet [49], CVSSN [51], DSGSF [47], GMA-Net [61], LMSS-NAS [60], GSCViT [62], DSFNet [53] and QuadNet [52].

The experimental results on the IP, KSC, and HH datasets are shown in Tables VI–VIII, where the best values for the classification accuracy of each category, as well as the best values for the OA, AA, and Kappa coefficients, are highlighted in bold. The results of these comparative experiments are illustrated in Figs. 6–8. As can be seen from Tables VI–VIII, the OA, AA, and Kappa coefficients of the proposed 3DAM-CAM DenseNet method are higher than those of the comparative methods on all three datasets, indicating that 3DAM-CAM DenseNet has a significant advantage in HSIC.

1) Quantitative Analysis: In the IP dataset, 3DAM-CAM DenseNet has an OA of 98.65%, which is higher than the deep learning methods ENL-FCN, A^{2}S^{2}K-ResNet, CVSSN, DSGSF, GMA-Net, GSCViT, DSFNet, and QuadNet with OA values of 0.84% and 0.05%, respectively, 0.96%, 1.33%, 13.39%, and 8.49%, and 0.95%, 2.02%, and 1.6%, respectively. In the KSC dataset, 3DAM-CAM DenseNet had an OA of 99.64%, and the OA values were 0.43%, 0.11%, 0.84%, 2.08%, 0.88%, 1.50%, 5.23%, 4.02%, and 1.72% higher, respectively, compared with the comparison methods. In the HH dataset, the OA of 3DAM-CAM DenseNet was 98.73% and the OA values were 7.03%, 1.10%, 1.00%, 4.80%, 4.24%, 0.78%, 2.58%, 5.29%, and 0.21% higher compared with the comparison method.

In the IP dataset, categories such as Alfalfa, Grass-pasture-mowed, Oats, and Stone-SteelTowers have relatively small sample sizes of 36, 22, 100, and 73, respectively. The classification accuracies achieved by the proposed method for these four categories are 96.48%, 95.06%, 100%, and 96.63%. This shows that the method proposed in this article remains effective even when working with small sample sizes.

2) Qualitative Analysis: Comparisons of classification results in visualizations for different methods on IP, KSC, and HH datasets are depicted in Figs. 6–8, where (c)–(i) show the visualized classification results for each method, where the IP and KSC datasets were trained using 10% of the samples, while the HH dataset was trained using 3% of the samples.

In Fig. 6, the visualization results of 3DAM-CAM DenseNet on the IP dataset exhibit the least noise and the highest classification accuracy, outperforming other methods. To better analyze the visualization results, the visualization results of the Oats category on the IP dataset are circled using white boxes in this section and combined with Table VI, the classification accuracy of this category on 3DAM-CAM DenseNet is 100%, and the visualization results are consistent with the results of the classification accuracy. In the overall visualization results, the prediction maps of GMA-Net and LMSS-NAS show obvious noise and poor classification results. Combined with Table VI, the OA of these two methods is 85.26% and 90.16%, respectively.

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In the KSC dataset, the white boxes in Fig. 7(b)–(l) highlight the categories of Slash pine and Cabbage palm. Observing Fig. 7(i) and (j), the visualization generated by GSViT and DSFNet classification methods shows the most noise and the worst performance. Specifically, in the white box of the GSViT method, some Slash pins are incorrectly classified as Cabbage palm and Oak, and some Cabbage palm pins are incorrectly classified as Oak and Slash pin. Some Cabbage palm categories in the white box of the DSFNet method are incorrectly classified as Slash pine. In contrast, as shown in Fig. 7(c) and (d), the ENL-FCN and A^2S^2K-ResNet methods only display very few misclassification results within the white box. The 3DAM-CAM DenseNet proposed in this article has the best visualization effect.

The visualization of the classification results of the HH dataset is shown in Fig. 8, with white boxes representing the Cotton firewood category in Fig. 8(b)–(j). By comparing the Cotton firewood categories in the white box, ENL-FCN and DSFNet methods exhibit the highest levels of noise in visualization. Specifically, in the white box of ENL-FCN, the Cotton firewood category is incorrectly classified as Cotton, Baresol, Whiter adish, and other categories. In the white box of the DSFNet method, the Cotton firewood category is incorrectly predicted as Red roof, Road, Cotton, and other categories. As shown in Fig. 8(d) and (h), the LMSS-NAS and QuadNet methods generate relatively less visualization noise on the HH dataset. According to Table VIII, the classification performance of these two types of OA is 97.95% and 98.52%, respectively, second only to the method proposed in this article. Through visual comparison, the 3DAM-CAM DenseNet method has the lowest noise and the best visualization results compared with the comparison method.

The 3DAM-CAM DenseNet method demonstrates significant advantages through both quantitative and visual analyses. This is attributed to its ability to construct the dependency relationships between the spatial and spectral information of the original HSI, as well as the interdependencies among pixel points across the three dimensions of spatial, spectral, and channel information. Consequently, crucial pixel-level information has a major role in classification. Additionally, the application of CAM allows DenseNet to emphasize the varying importance of features across different channels.

Furthermore, this article further evaluates the performance of 3DAM-CAM DenseNet under various training sample ratios, with ratios of 2%, 3%, 5%, 8%, and 10% for the IP and KSC datasets, and 0.5%, 1%, 1.5%, 2%, 2.5%, and 3% for the HH dataset. Fig. 9 illustrates the OA values of the IP, KSC, and HH datasets under different training sample ratios. The proposed method demonstrates significant advantages in maintaining high OA in the IP and KSC datasets in a wide range of training ratios, from 2% to 10%. Similarly, it outperforms the baseline methods in the HH dataset with training ratios ranging from 0.5% to 3%. Furthermore, our method exhibits strong stability and adaptability across all three datasets as the proportion of the training sample increases. These results indicate that the model is not only robust under data scarcity but also effectively leverages richer training data to enhance performance, thereby showcasing its superior generalization capability and classification accuracy.

E. Ablation Experiment

To analyze the impact of the two branches on the network model's performance, we modified the ablation experiments. The findings are detailed in Table IX. The baseline network was redefined as Conv(1,1,7) + BN + DenseNet. In Table IX, the "✓" signifies that the corresponding module has been integrated into the baseline network.

When only CAM was added to the baseline network, the connection between different channel features was established, highlighting the importance of using channel features. For the three datasets, IP, KSC, and HH, the OA values increased by 0.77%, 0.30%, and 0.97%, respectively, compared with the baseline network.

When only 3DAM-SS was added, the relationship between spatial and spectral information in the original data was established. For the three datasets, IP, KSC, and HH, the OA values increased by 0.50%, 0.20%, and 0.89%, respectively, compared with the baseline network. When only 3DAM-SSC was added, the interaction between feature pixels in the spatial–spectral channel dimensions was established, leading to an increase in OA value of 0.40%, 0.17%, and 0.74% for the IP, KSC, and HH datasets, respectively, compared with the baseline network.

When both 3DAM-SS and 3DAM-SSC were added to the baseline network, the OA values increased by 0.33%, 0.06%, and 0.43% for the IP, KSC, and HH datasets, respectively. When the proposed model added 3DAM-SS, 3DAM-SSC, and CAM simultaneously, the OA values increased by 0.71%, 0.78%, and 0.47% for the IP, KSC, and HH datasets, respectively, compared with the baseline network.

When we removed the Conv(3, 3, 7) branch from the proposed model, the OA values for the IP, KSC, and HH datasets all decreased. This suggests that the joint use of spatial and spectral features helps to improve the model's classification performance.

The analysis and results in the figures above confirm that the design of each module in the proposed method is reasonable.

F. Comparative Experiments Based on Different Attentional Mechanisms

To verify the performance differences and computational complexity of various AMs under identical conditions, this article use the ResNet based on 3DCNN as the base network. CBAM [45], nonlocal [55], quadlet attention [52], SSAM [53], and 3DAM-SSC were, respectively, embedded into the same position of the ResNet. The results of the comparative experiments and complexity evaluations are presented in Table X. Table X shows the classification results of each of the five AMs combined with ResNet, and the results show that all five AMs contribute to the classification performance. On the IP dataset, the OA values of the five models outperform ResNet by 1.12%, 0.89%, 0.80%, 0.62%, and 1.16%, respectively.

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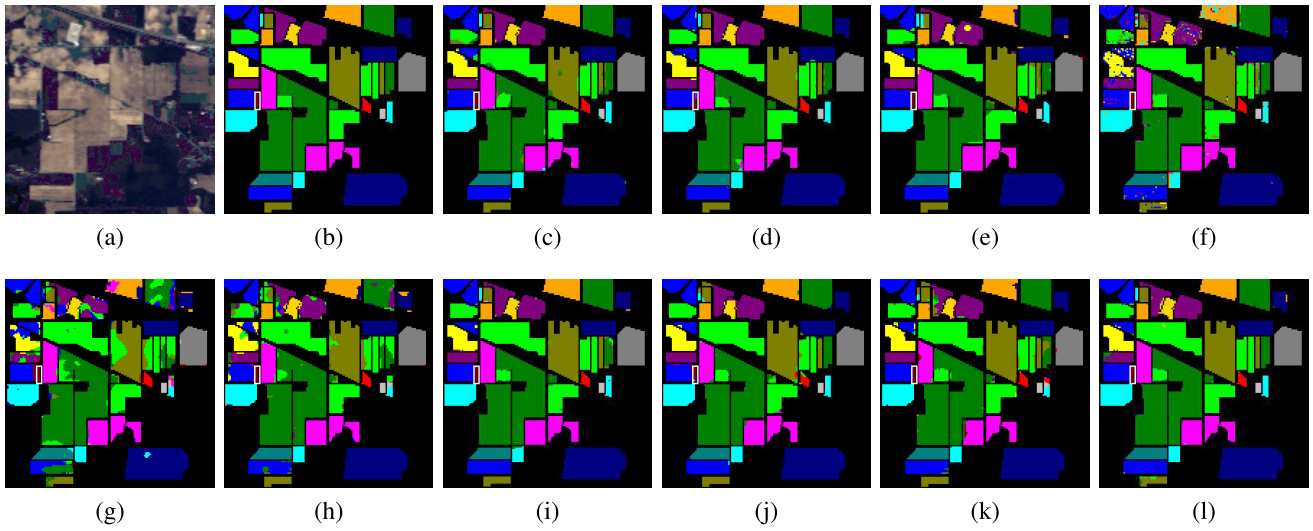


Fig. 6. Classification maps for the IP dataset. (a) False-color image (10, 20, 30). (b) Ground-truth map. (c) ENL-FCN. (d) A^2S^2K-ResNet. (e) CVSSN. (f) DSGSF. (g) GMA-Net. (h) LMSS-NAS. (i) GSCViT. (j) DSFNet. (k) QuadNet. (l) 3DAM-CAM DenseNet.

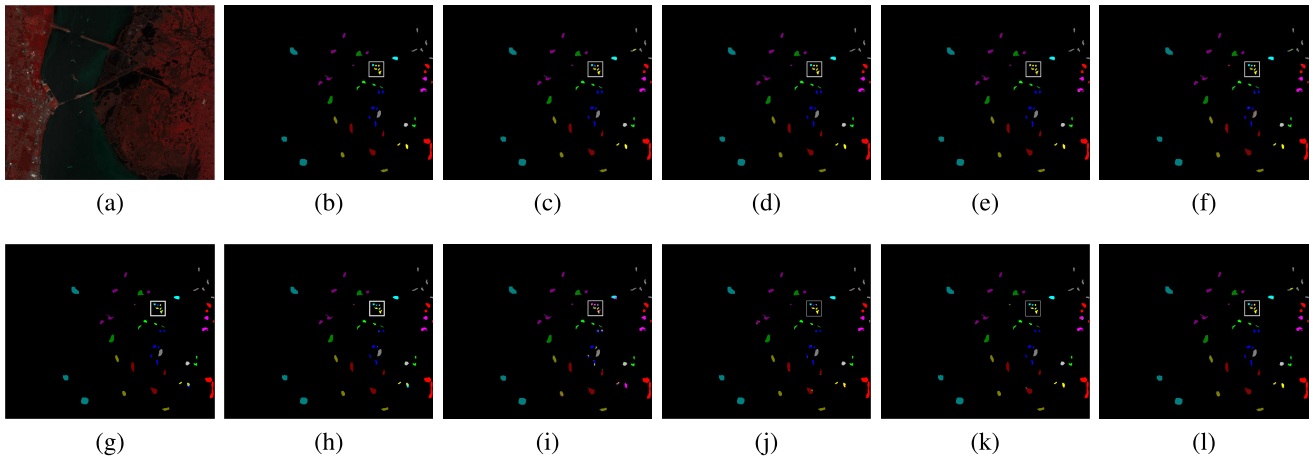


Fig. 7. Classification maps for the KSC dataset. (a) False-color image (10, 20, 30). (b) Ground-truth map. (c) ENL-FCN. (d) A^2S^2K-ResNet. (e) CVSSN. (f) DSGSF. (g) GMA-Net. (h) LMSS-NAS. (i) GSCViT. (j) DSFNet. (k) QuadNet. (l) 3DAM-CAM DenseNet.

TABLE VII
OA, AA, AND KAPPA VALUES ON THE KSC DATASET USING FIXED 10% OF TRAINING SAMPLES

Class	Methods									
#	ENL-FCN	A^2S^2K-ResNet	CVSSN	DSGSF	GMA-Net	LMSS-NAS	GSCViT	DSFNet	QuadNet	Proposed
1	98.96±0.015	**100.00±0.000**	**100±0.000**	97.91±0.013	**100±0.000**	**100±0.000**	96.48±0.049	95.11±0.008	99.67±0.005	**100.00±0.000**
2	**100.00±0.000**	97.03±0.042	97.44±0.910	95.83±0.023	**100±0.000**	**100±0.000**	95.79±0.054	95.84±0.018	96.18±0.024	98.21±0.022
3	98.73±0.012	**100.00±0.000**	95.55±0.182	98.70±0.015	**100±0.000**	**100±0.000**	96.35±0.036	87.02±0.035	99.32±0.006	99.65±0.0049
4	92.43±0.093	**100.00±0.000**	92.02±0.364	97.15±0.025	99.14±0.014	98.15±0.012	68.02±0.060	88.77±0.042	93.63±0.053	97.16±0.040
5	95.35±0.045	93.78±0.050	97.09±0.206	93.26±0.051	94.33±0.022	92.91±0.017	75.66±0.163	94.63±0.051	84.38±0.112	**100.00±0.000**
6	**100.00±0.000**	**100.00±0.000**	94.64±0.383	95.68±0.034	**100±0.000**	73.58±0.146	65.79±0.221	91.99±0.053	97.62±0.030	**100.00±0.000**
7	99.21±0.011	99.23±0.005	**100.00±0.000**	97.64±0.016	**100±0.000**	**100±0.000**	66.55±0.246	87.55±0.049	**100.00±0.000**	**100.00±0.000**
8	**100.00±0.000**	99.90±0.001	**100.00±0.000**	96.75±0.018	**100±0.000**	98.28±0.037	99.09±0.007	95.76±0.013	97.04±0.016	99.80±0.001
9	**100.00±0.000**	99.92±0.001	**100.00±0.000**	95.87±0.039	89.40±0.016	**100±0.000**	98.27±0.019	94.37±0.031	**100.00±0.000**	**100.00±0.000**
10	**100.00±0.000**	99.18±0.001	**100.00±0.000**	97.20±0.022	**100±0.000**	**100±0.000**	86.78±0.187	96.33±0.017	97.12±0.021	98.88±0.010
11	99.51±0.007	99.90±0.001	99.04±0.136	97.70±0.016	**100±0.000**	**100±0.000**	**100.00±0.000**	98.36±0.013	**100±0.000**	**100.00±0.000**
12	99.83±0.001	**100.00±0.000**	99.63±0.520	98.72±0.004	**100±0.000**	**100±0.000**	94.85±0.026	98.89±0.010	97.20±0.040	**100.00±0.000**
13	**100.00±0.000**	**100.00±0.000**	**100.00±0.000**	99.82±0.001	**100±0.000**	**100±0.000**	99.91±0.001	99.95±0.006	**100.00±0.000**	**100.00±0.000**
OA (%)	99.21±0.005	99.53±0.002	98.80±0.270	97.56±0.004	98.73±0.014	98.11±0.016	94.41±0.013	95.62±0.001	97.92±0.003	**99.64±0.003**
AA (%)	98.77±0.006	99.15±0.005	98.11±0.420	97.09±0.006	98.68±0.006	97.14±0.021	91.46±0.016	94.20±0.006	97.09±0.001	**99.52±0.004**
Kappa × 100	99.12±0.007	99.48±0.002	98.67±0.300	97.28±0.004	98.58±0.010	97.89±0.016	93.77±0.014	95.13±0.001	97.68±0.004	**99.60±0.003**

On the KSC dataset, the OA values of the five models outperform ResNet by 0.84%, 0.21%, 0.14%, 1.10%, and 1.29%, respectively. On the HH dataset, the OA values of the five models outperform ResNet by 1.16%, 0.99%, 0.32%, 0.89%, and 1.20%, respectively. The above results show that the classification performance of ResNet + 3DAM-SSC is optimal, indicating that 3DAM-SSC utilizes 3-D pixel information to establish weights in a way that helps to improve the model classification performance. However, accordingly, the floating point operation of ResNet + 3DAM-SSC is 5.57

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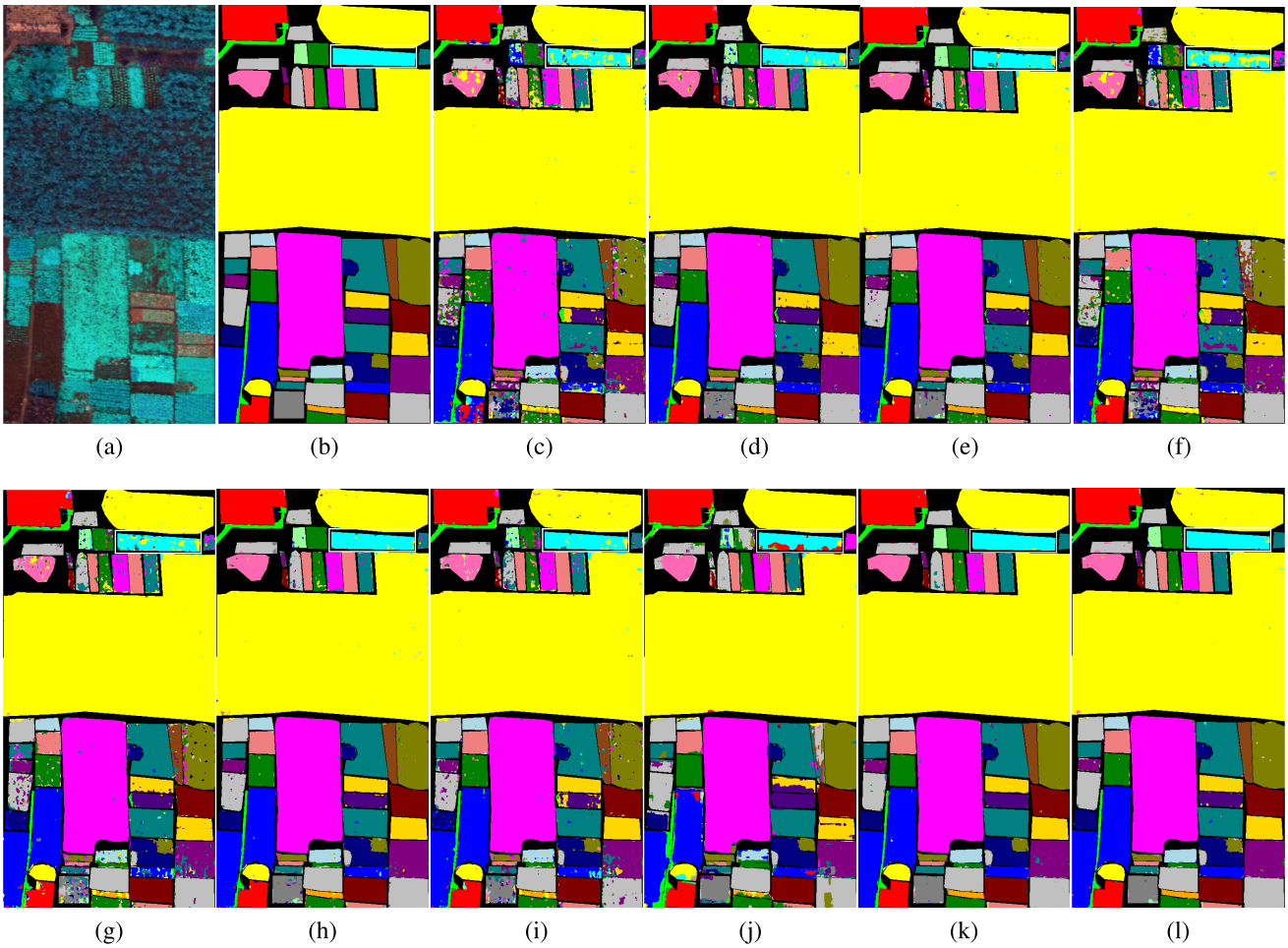


Fig. 8. Classification maps for the HH dataset. (a) False-color image (21, 41, 121). (b) Ground-truth map. (c) ENL-FCN. (d) A^2S^2K-ResNet. (e) CVSSN. (f) DSGSF. (g) GMA-Net. (h) LMSS-NAS. (i) GSCViT. (j) DSFNet. (k) QuadNet. (l) 3DAM-CAM DenseNet.

TABLE VIII

OA, AA, AND KAPPA VALUES ON THE HH DATASET USING FIXED 3% OF TRAINING SAMPLES

Class	Methods									
#	ENL-FCN	A^2S^2K-ResNet	CVSSN	DSGSF	GMA-Net	LMSS-NAS	GSCViT	DSFNet	QuadNet	Proposed
1	96.41±0.02	98.03±0.01	**98.63±0.00**	97.97±0.00	97.56±0.00	**99.14±0.00**	98.04±0.00	89.90±0.061	99.54±0.002	**99.42±0.00**
2	79.15±0.00	92.06±0.02	93.23±0.00	88.18±0.02	83.62±0.03	95.39±0.01	87.17±0.01	61.13±0.098	97.13±0.011	**97.02±0.02**
3	88.36±0.02	97.67±0.00	**98.13±0.00**	92.26±0.03	93.88±0.02	97.25±0.01	96.38±0.01	90.71±0.043	95.92±0.025	**98.57±0.00**
4	97.28±0.00	**99.43±0.00**	**99.55±0.00**	97.87±0.01	98.40±0.00	99.28±0.00	99.23±0.00	99.61±0.002	99.87±0.001	**99.79±0.00**
5	81.65±0.00	95.49±0.02	93.58±0.02	88.76±0.01	88.21±0.04	95.13±0.01	86.89±0.02	88.69±0.045	94.07±0.049	94.92±0.01
6	95.47±0.00	98.72±0.00	98.99±0.00	98.05±0.01	97.28±0.01	98.84±0.00	98.02±0.01	98.03±0.008	99.40±0.003	**99.57±0.00**
7	88.72±0.01	96.78±0.01	95.75±0.01	88.39±0.04	89.33±0.01	97.17±0.01	94.01±0.01	86.37±0.029	97.80±0.007	**97.49±0.01**
8	74.42±0.04	86.73±0.02	89.61±0.00	74.96±0.04	82.57±0.05	90.32±0.01	83.36±0.00	78.66±0.010	95.73±0.004	**94.93±0.00**
9	98.66±0.01	**99.24±0.00**	98.54±0.00	99.25±0.01	98.35±0.00	99.42±0.00	98.28±0.00	89.46±0.116	99.54±0.003	**99.43±0.00**
10	91.07±0.01	96.90±0.01	98.37±0.00	89.30±0.07	95.43±0.00	98.96±0.00	96.94±0.00	86.51±0.036	98.80±0.012	**99.32±0.00**
11	73.75±0.03	93.68±0.00	95.22±0.01	81.31±0.03	82.57±0.01	93.39±0.01	90.02±0.03	86.21±0.040	96.44±0.006	**96.75±0.01**
12	73.57±0.04	91.91±0.00	90.99±0.02	76.67±0.10	80.20±0.01	92.53±0.02	84.42±0.04	88.01±0.019	90.86±0.083	**92.52±0.06**
13	78.23±0.05	95.46±0.01	94.15±0.00	86.59±0.02	84.37±0.01	95.81±0.00	88.96±0.02	88.38±0.029	97.55±0.016	**98.54±0.00**
14	89.57±0.06	98.08±0.00	98.50±0.00	93.44±0.04	92.89±0.00	98.79±0.00	97.05±0.00	90.97±0.041	98.70±0.001	**98.67±0.00**
15	95.19±0.01	87.24±0.09	97.43±0.01	94.63±0.00	90.51±0.01	91.30±0.056	96.46±0.021	91.30±0.056	96.46±0.021	**93.02±0.01**
16	88.40±0.00	98.32±0.00	98.54±0.01	92.24±0.03	96.76±0.02	97.81±0.00	92.30±0.06	83.72±0.081	99.23±0.005	**99.05±0.00**
17	84.77±0.02	92.04±0.02	93.94±0.01	83.13±0.03	87.54±0.01	95.27±0.00	88.24±0.01	83.86±0.206	95.01±0.041	**99.80±0.00**
18	80.51±0.08	93.24±0.02	91.73±0.05	89.57±0.03	91.47±0.03	95.53±0.02	97.12±0.01	98.35±0.016	94.74±0.024	96.08±0.01
19	86.15±0.01	93.17±0.04	94.97±0.03	96.96±0.01	94.93±0.01	95.97±0.01	93.70±0.03	90.89±0.039	98.01±0.006	**98.42±0.01**
20	82.98±0.07	89.98±0.01	92.30±0.03	73.36±0.02	84.62±0.08	94.67±0.03	94.53±0.00	89.98±0.058	98.16±0.011	**95.64±0.00**
21	65.95±0.00	93.29±0.00	89.04±0.05	36.65±0.30	69.53±0.09	92.03±0.05	81.56±0.09	69.89±0.249	95.25±0.046	**94.38±0.01**
22	84.85±0.02	94.90±0.01	93.74±0.02	88.69±0.03	89.94±0.01	95.64±0.02	92.56±0.01	92.36±0.051	97.80±0.016	**90.63±0.01**
OA(%)	91.70±0.87	97.63±0.02	97.73±0.01	93.93±0.01	94.49±0.02	97.95±0.03	96.15±0.04	93.44±0.008	98.52±0.003	**98.73±0.02**
AA(%)	85.03±0.02	94.96±0.04	95.12±0.05	86.40±0.04	89.86±0.02	96.04±0.01	92.24±0.02	87.41±0.033	97.09±0.005	**96.70±0.03**
Kappa×100	89.46±0.01	97.01±0.02	97.13±0.01	92.17±0.08	93.01±0.01	97.41±0.04	95.13±0.05	91.71±0.011	98.13±0.016	**98.40±0.02**

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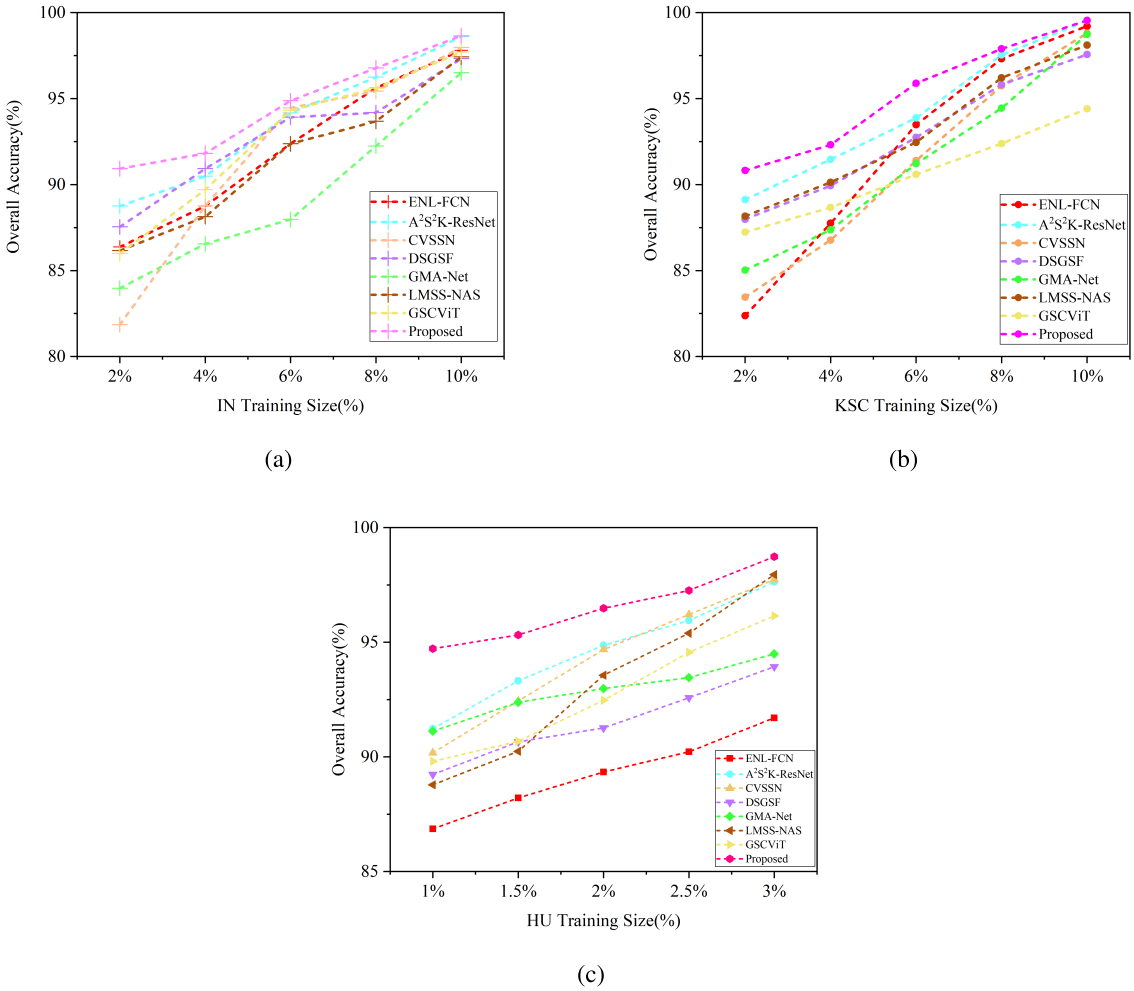


Fig. 9. OA of 3DAM-CAM DenseNet on different datasets with different training set scales.(a) IP, (b) KSC, and (c) HH data sets.

TABLE IX
ACCURACY ANALYSIS IN TERMS OF OA FOR DIFFERENT MODULES OF THE PROPOSED FRAMEWORK

	Models				OA (%)		
	CAM	3DAM-SS	3DAM-SSC	Conv(3,3,7)	IP	KSC	HH
1	✗	✗	✗	✗	96.83	98.28	96.77
2	✗	✗	✗	✓	97.31	99.06	97.29
3	✓	✗	✗	✓	97.88	99.34	97.76
4	✗	✓	✗	✓	98.15	99.44	97.84
5	✗	✗	✓	✓	98.25	99.47	97.99
6	✗	✓	✓	✓	98.32	99.58	98.30
7	✓	✓	✓	✓	**98.65**	**99.64**	**98.73**

(10^9 × FLOPs), which is the highest compared with other AMs combined with ResNet, and the number of parameters is basically the same as other models.

G. Analysis of Computational Complexity and Robustness

Table XI shows the complexity of the model of various networks, including floating-point operations (FLOPs), the number of parameters (Param), and the computation time per sample on three datasets. As shown in Table XI, GMA-Net uses the most parameters, while ENL-FCN and the proposed method require fewer parameters. DSGSF and GMA-Net have the highest computational costs, but our model also reaches 7.48 (10^9 × FLOPs), while CVSSN has the lowest computational cost. For the IP dataset, the proposed model takes 7.6 ms/sample. On the KSC dataset, A^2S^2K-ResNet is the fastest, taking only 0.5 ms/sample, while our method takes more time to process. In the HH dataset, CVSSN is the fastest, while our method reaches 15.5 ms/sample.

To evaluate the noise resistance and robustness of our proposed model, we add different levels of Gaussian noise to the three original datasets. We add Gaussian noise with the same intensity $N(0, \sigma^2)$ (mean zero and variance σ^2) to each band in the dataset, and the noise is independent and distributed identically (i.i.d.). Table XII shows the classification results of 3DAM-CAM DenseNet with different Gaussian noise added to the three datasets. Through Table XII, it can be seen that as the standard deviation of Gaussian noise increases, the OA, AA, and Kappa values of 3DAM-CAM DenseNet on the three datasets of IN, KSC, and HH are also reduced, and when the standard deviation of Gaussian noise is 0.08, the OA of IN, KSC, and HH are 97.02%, 92.00%, and 97.27%, respectively, and the classification performances

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TABLE X
COMPARISON OF DIFFERENT AMS EMBEDDED INTO RESNET UNDER THE SAME CONDITIONS

Dataset	Matrices	ResNet	ResNet+CBAM	ResNet+Non-local	ResNet+Quadlet	ResNet+SSAM	ResNet+3DAM-SSC
IP	OA(%)	93.59 ± 0.019	94.71 ± 0.012	94.48 ± 0.024	94.39 ± 0.005	94.21 ± 0.017	**94.75 ± 0.005**
	AA(%)	91.58 ± 0.021	91.42 ± 0.018	91.83 ± 0.028	91.68 ± 0.030	90.73 ± 0.023	**92.26 ± 0.018**
	Kappa×100	92.71 ± 0.021	93.96 ± 0.014	93.70 ± 0.017	93.61 ± 0.008	93.39 ± 0.012	**94.01 ± 0.006**
KSC	OA(%)	96.36 ± 0.006	97.20 ± 0.026	96.57 ± 0.012	96.50 ± 0.007	97.46 ± 0.009	**97.65 ± 0.013**
	AA(%)	94.43 ± 0.008	95.09 ± 0.028	94.22 ± 0.016	95.12 ± 0.012	95.82 ± 0.015	**96.26 ± 0.017**
	Kappa×100	95.96 ± 0.007	96.88 ± 0.029	96.18 ± 0.013	96.11 ± 0.008	97.40 ± 0.016	**97.39 ± 0.010**
HH	OA(%)	96.65 ± 0.008	97.81 ± 0.006	97.64 ± 0.002	96.97 ± 0.007	97.54 ± 0.005	**97.85 ± 0.003**
	AA(%)	93.29 ± 0.017	96.21 ± 0.014	94.99 ± 0.016	94.11 ± 0.016	94.16 ± 0.019	**95.67 ± 0.006**
	Kappa×100	95.76 ± 0.010	97.23 ± 0.008	97.01 ± 0.003	96.68 ± 0.009	96.89 ± 0.006	**97.28 ± 0.004**
	Params (M)	3.17	3.17	3.18	3.18	3.17	3.17
	Computation Cost (10^9 FLOPs)	5.32	5.32	5.46	5.36	5.32	5.57

TABLE XI
COMPARISON OF COMPUTATIONAL COMPLEXITY OF METHODS ON THREE DATASETS

	ENL-FCN	A^2S^2K-ResNet	DSGSF	CVSSN	GMA-Net	proposed
Params	19.3K	480.6K	410.5K	270.4K	1.6M	461.7K
Computation Cost (10^9 × FLOPs)	3.64	3.72	9.34	0.35	6.75	7.48
Datasets	Computation Time (ms/sample)					
IP	2.7	3.9	1.6	2.4	3.1	7.6
KSC	1.3	0.5	1.1	0.3	0.8	1.6
HH	5.3	7.6	18.4	0.8	10.4	15.5

TABLE XII
CLASSIFICATION PERFORMANCE WITH DIFFERENT LEVELS OF GAUSSIAN ERROR ADDED TO IN, KSC, AND HH DATASETS

σ	IN			KSC			HH		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
0	98.65	98.19	98.46	99.64	99.52	99.60	98.73	96.70	98.40
0.03	97.32	93.88	96.95	95.60	94.46	95.10	97.89	94.11	97.34
0.05	97.35	91.31	96.78	94.98	94.41	94.41	97.37	93.24	97.18
0.08	97.02	89.65	96.61	92.00	93.37	91.11	97.27	93.78	96.55

are still considerable, which indicates that the 3DAM-CAM DenseNet has a certain suppression effect on noise.

IV. CONCLUSION

To further improve the performance of HSI and highlight different roles of different pixels and channel features in classification, this article proposes an HSIC method with 3-D pixel-point dependencies and channel feature dependencies. Comparison experiments show that the classification effect of the method proposed in this article is better than that of other methods and the method could still achieve better results for small-sample training. In addition, the results of ablation experiments verify that the design of this method is reasonable.

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